

ECTE 250

DETAILED DESIGN REPORT DELIVERABLE 3

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Table Of Contents

1. Introduction.....	3
2. Overview of Inputs.....	3
3. State Machine.....	4
3.1 State Diagram.....	5
3.2 States.....	6
3.3 State Table.....	11
3.4 State Expressions.....	11
3.5 555 Timer Circuit.....	12
3.6 Clocking Oscillations.....	12
3.7 Power Circuit.....	14
3.8 Motor Circuit.....	15
3.9 Debouncing Circuit.....	16
4. Sensor Circuit.....	17
4.1 Camera (LDR).....	18
4.2 Gas sensor.....	19
4.3 Flowchart of Sensor Logic.....	20
4.4 Block Chart Diagram.....	21
5 Prototype Sketch.....	22
6. Feedback.....	23
8. Conclusion.....	25

1. Introduction

This report presents the Design Simulation for ECTE250 Deliverable 3, showing how the prototype's main subsystems work together to achieve the required functionality. Through simulation in Multisim, Tinkercad, and related tools, the circuit behaviour, state transitions, and output responses were verified, confirming the design is ready for physical prototyping and later lab testing.

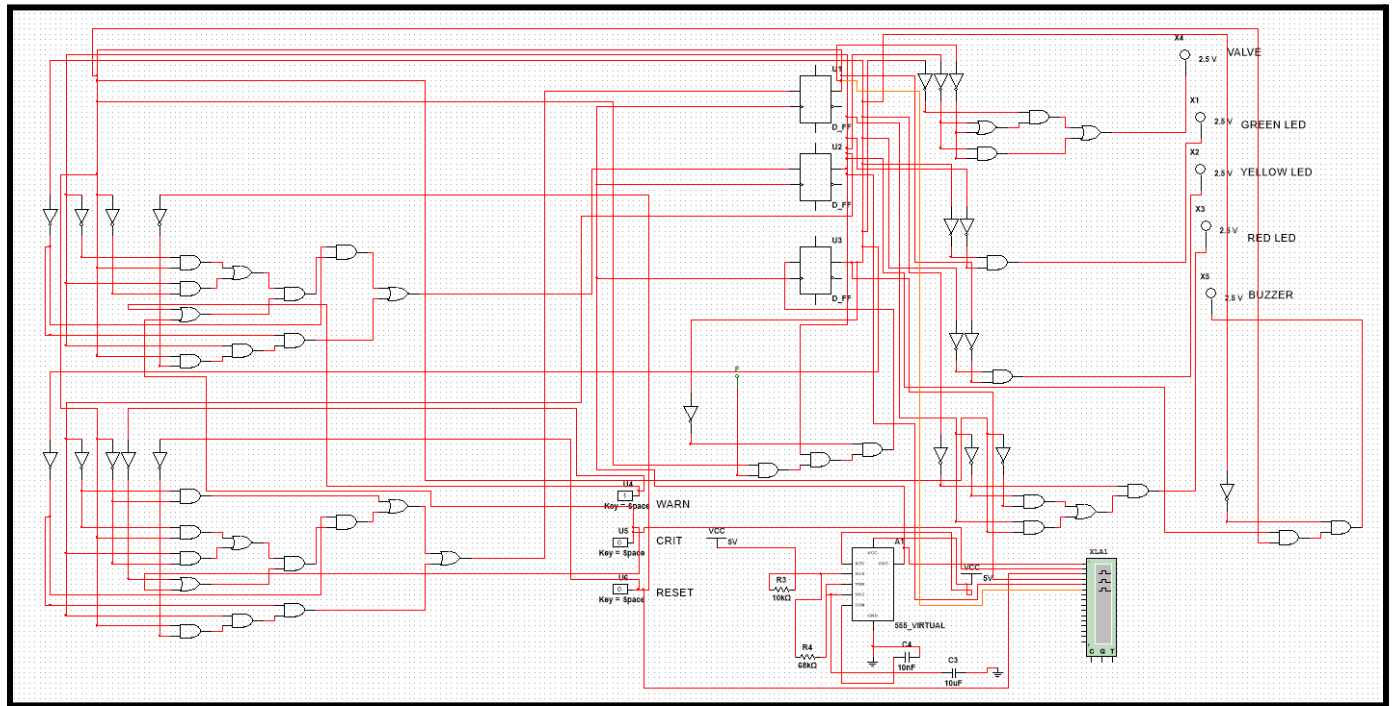
2. Overview of Inputs

WARN - This input signal comes from a camera sensor (which is linked to the Jetson Nano), when it detects low levels of contaminants/nurdles. It essentially means the system has identified a potential hazard that requires caution

CRIT- This input signal comes when the gas sensor detects hazardous gas leaks or if the camera sensor detects a high density of nurdles/contaminants.It represents a confirmed emergency that requires the system to immediately close the valve and trigger alarms

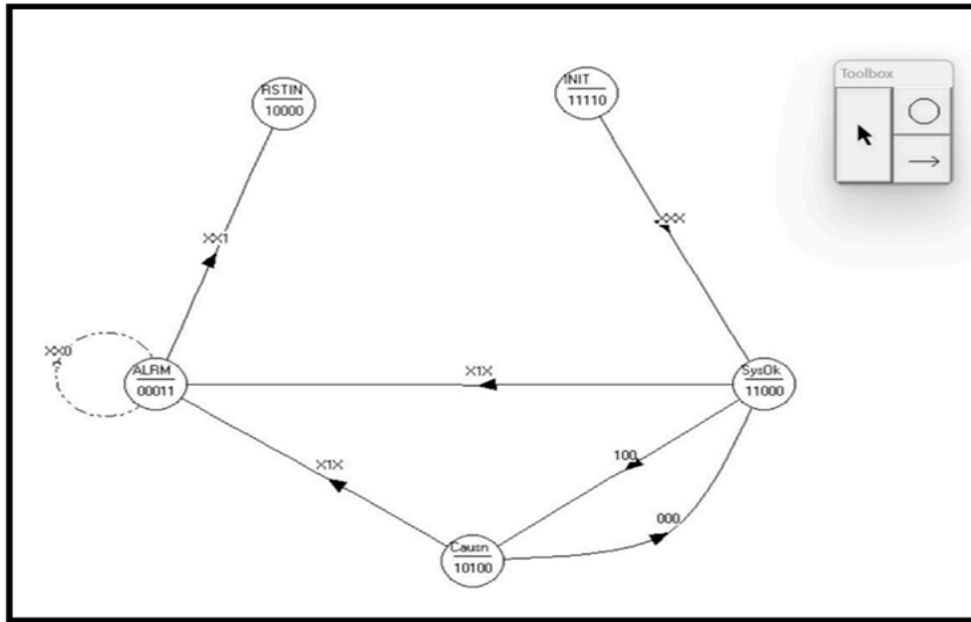
RST-This input is controlled by a manual push button that must be physically pressed by an operator. It signifies that the hazard has been acknowledged and is the only way to return the system from the critical state to its normal state.

3. State Machine



The circuit was built in Multisim based on our initial state machine logic (obtained from Bool-Duesto). It uses three D flip-flops to act as the memory for our three-bit state. We used a variety of AND, OR, and NOT gates to process the inputs—WARN, CRIT, and RESET—and decide what the next state should be. These same gates also control our five outputs: the three status LEDs, the buzzer, and the main valve.

3.1 State Diagram

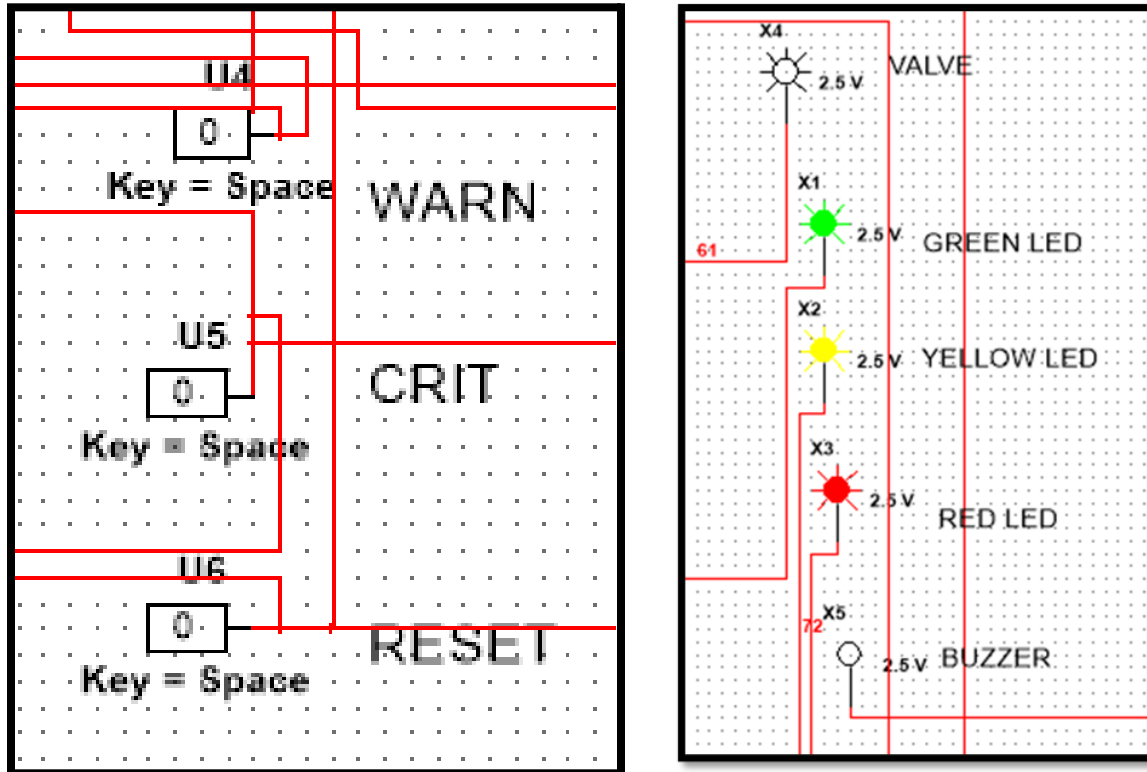


The state machine for our system has 5 states, which are described below in their respective order;

- S0 - INIT (Initialization State)
- S1 - SysOk (Normal State)
- S2 - Causn (Warning State)
- S3 - ALRM (Critical State)
- S4 - RSTIN(Reset initialization)

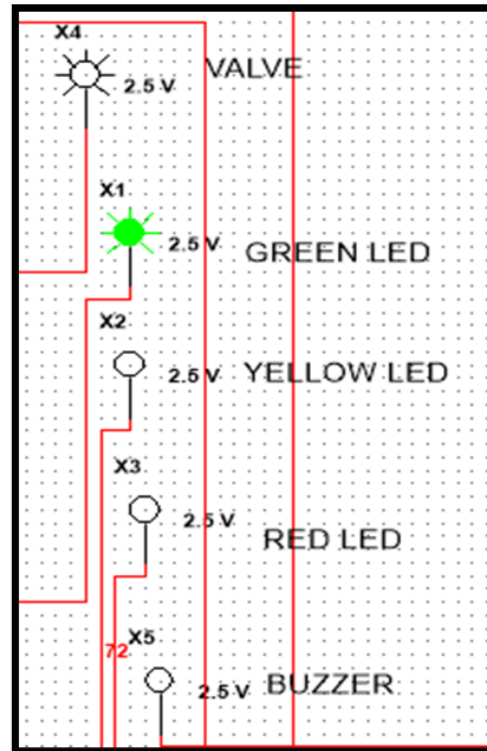
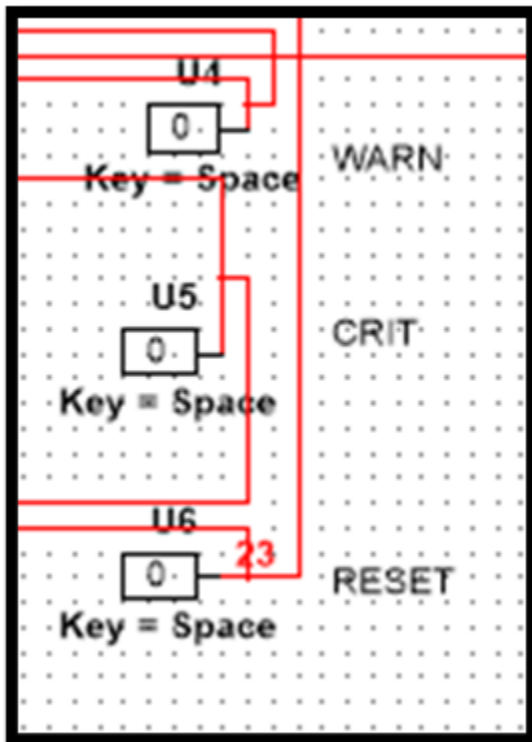
3.2 States

1.) S0 - INIT



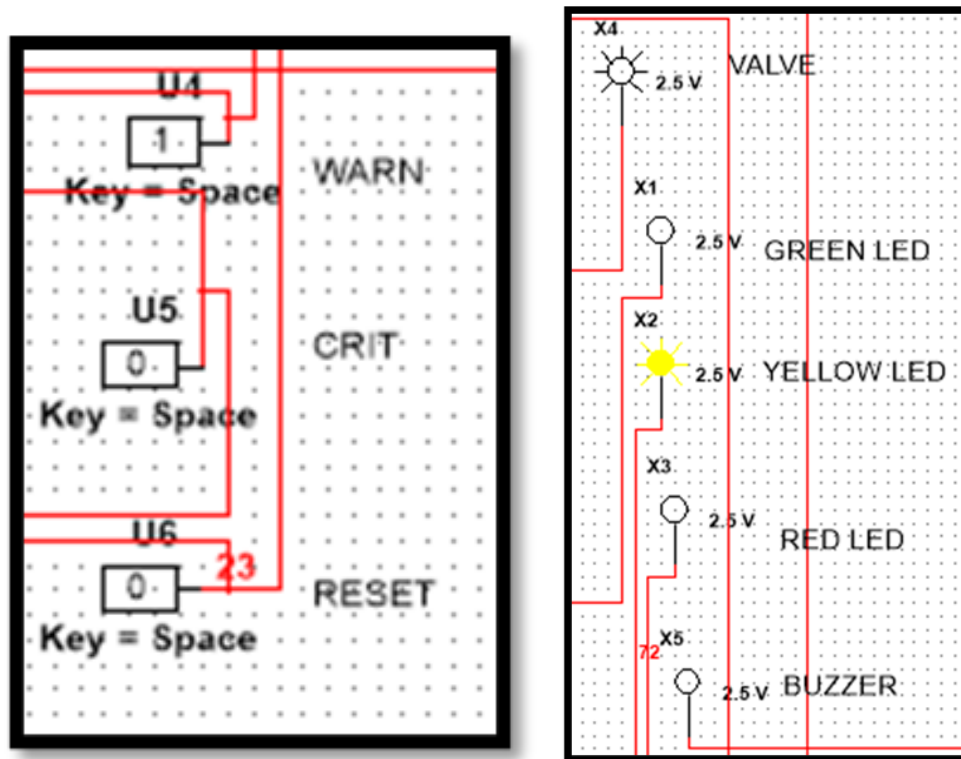
This is the FSM power-up state, activated as soon as the system is powered on. In this stage, the inputs are 000 and the outputs are 11110, turning on all LEDs while keeping the valve open as a self-test before the system shifts to normal operation on the first clock pulse. Once the first clock pulse from the 555 timer occurs, the logic moves the system from this state into the normal operating mode

2.)S1 - SyskOk



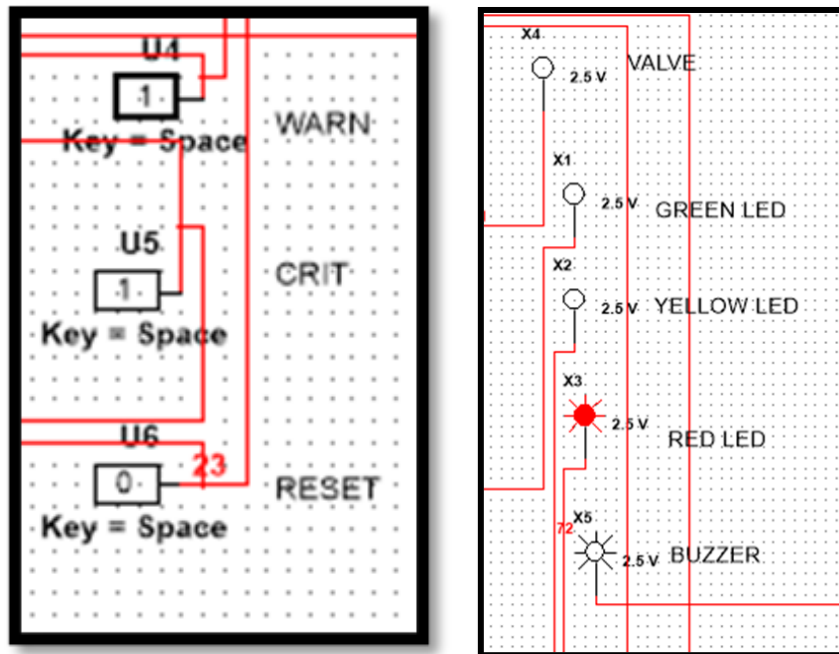
This is the baseline safe mode for the Nurdle DNA system. It is reached when the sensors detect no contaminants or gas leaks, meaning the inputs are 000. In this state, the outputs are 11000, which means the valve is open and the green LED is on. This indicates that the environment is safe and everything is running correctly.

3.)S2 -Causn



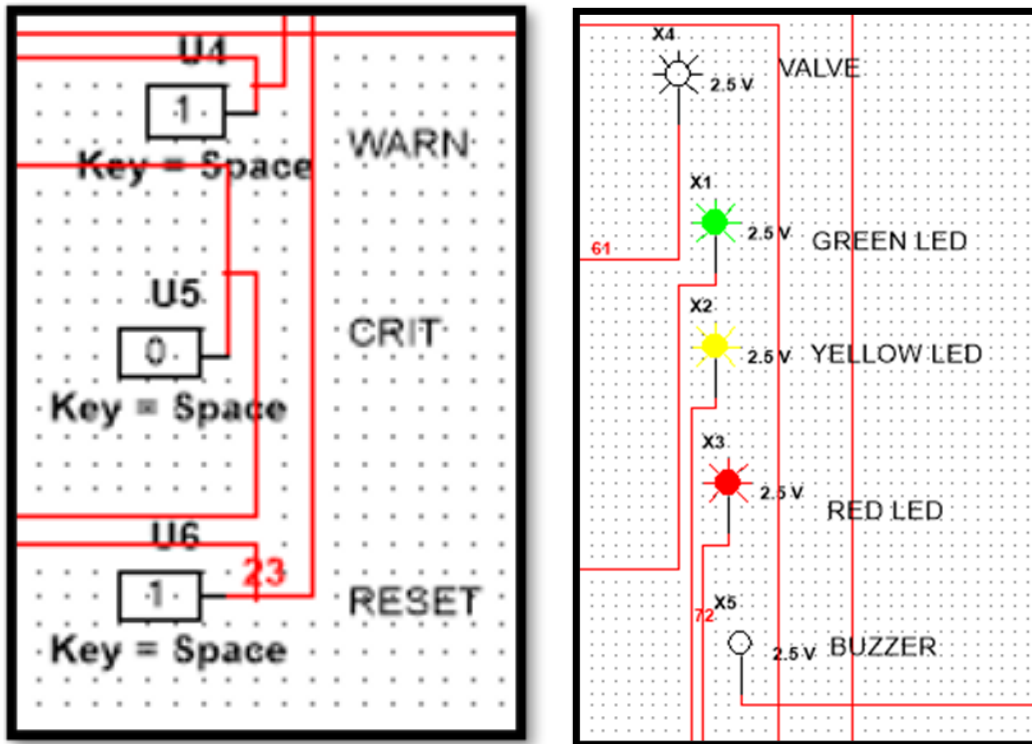
The system moves to this state when a low-level contaminants are detected. It is reached when the WARN input is triggered, making the inputs 100. The outputs change to 10100, which turns off the green LED and turns on the yellow LED. The valve stays open so the system can keep working while the operator is alerted to the potential problem.

4.) S3 -ALRM



This is the alarm state reached when high contaminant level or gas leak is detected. It is triggered by the CRIT sensor, with inputs of X10. The outputs become 00011, which immediately closes the valve and activates the red LED and buzzer. This is a safety lock state, meaning the alarm stays active until someone manually resets the system.

5.) S4 -RSTIN



This state is reached when a person presses the reset button to clear an alarm. It happens whenever the RESET input is 1 (XX1). The outputs are 11110, which reopens the valve and clears the light and sound indicators. This allows the system to cycle back to the normal state, S1.

3.3 State Table

	000	001	010	011	100	101	110	111
SysOk	X	X	ALRM	ALRM	Causn	X	ALRM	ALRM
Causn	SysOk	X	ALRM	ALRM	X	X	ALRM	ALRM
ALRM	ALRM	RSTIN	ALRM	RSTIN	ALRM	RSTIN	ALRM	RSTIN
RSTIN	X	X	X	X	X	X	X	X
INIT	SysOk	SysOk	SysOk	SysOk	SysOk	SysOk	SysOk	SysOk

	Qi	Output		000	001	010	011	100	101	110	111
SysOk	000	11000	000	xxx	xxx	010	010	001	xxx	010	010
Causn	001	10100	001	000	xxx	010	010	xxx	xxx	010	010
ALRM	010	00011	010	010	011	010	011	010	011	010	011
RSTIN	011	10000	011	xxx	xxx	xxx	xxx	xxx	xxx	xxx	xxx
INIT	100	11110	100	000	000	000	000	000	000	000	000
X	101	XXXXX	101	xxx	xxx	xxx	xxx	xxx	xxx	xxx	xxx
X	110	XXXXX	110	xxx	xxx	xxx	xxx	xxx	xxx	xxx	xxx
X	111	XXXXX	111	xxx	xxx	xxx	xxx	xxx	xxx	xxx	xxx

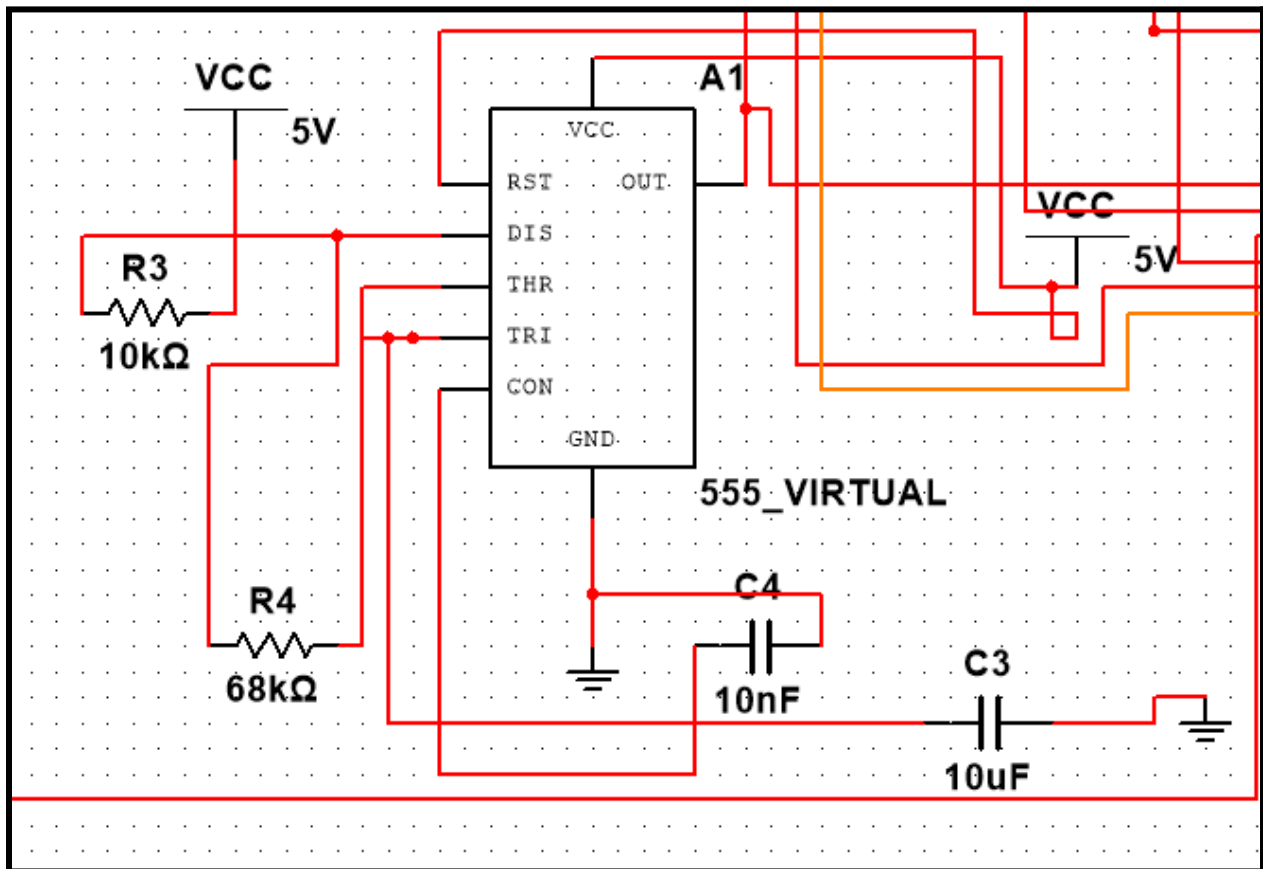
3.4 State Expressions

```

J0=(~CRIT*~q2*~q1)+(WARN*q1)
K0=(1)
J1=(CRIT*~q2)
K1=(0)
J2=(0)
K2=(1)
D0=(WARN*q1)+(~CRIT*~q2*~q1*~q0)
D1=(q1)+(CRIT*~q2)
D2=(0)
VALV=(q1*~q0)
LedG=(q1*~q0)+(q2)
LedY=(~q1*q0)+(q2)
LedR=(~q1*~q0)
BUZZ=(~q1)+(q0)

```

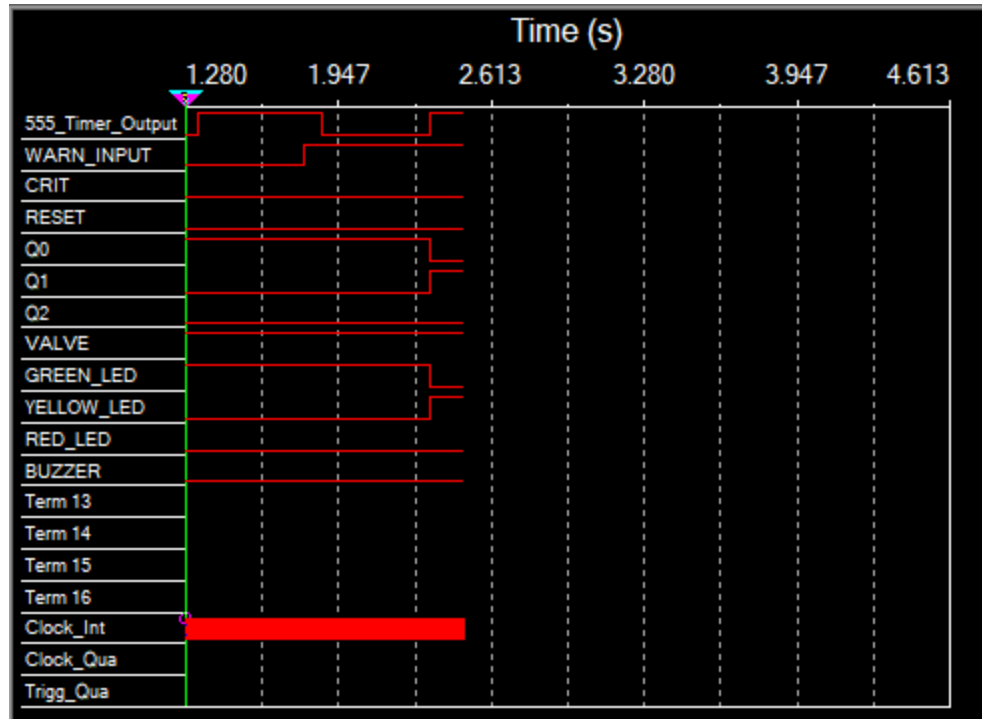
3.5 555 Timer Circuit



To ensure synchronous operation, a 555 Timer IC was integrated into the circuit to function as a standalone hardware clock. In this design, the 555 Timer is configured in astable mode. We used a 10 kΩ resistor (R_1), a 68kΩ resistor (R_2), and a 10 μF capacitor (C_1) to set the timing. These specific values create a 1 Hz pulse, meaning the machine updates exactly once per second.

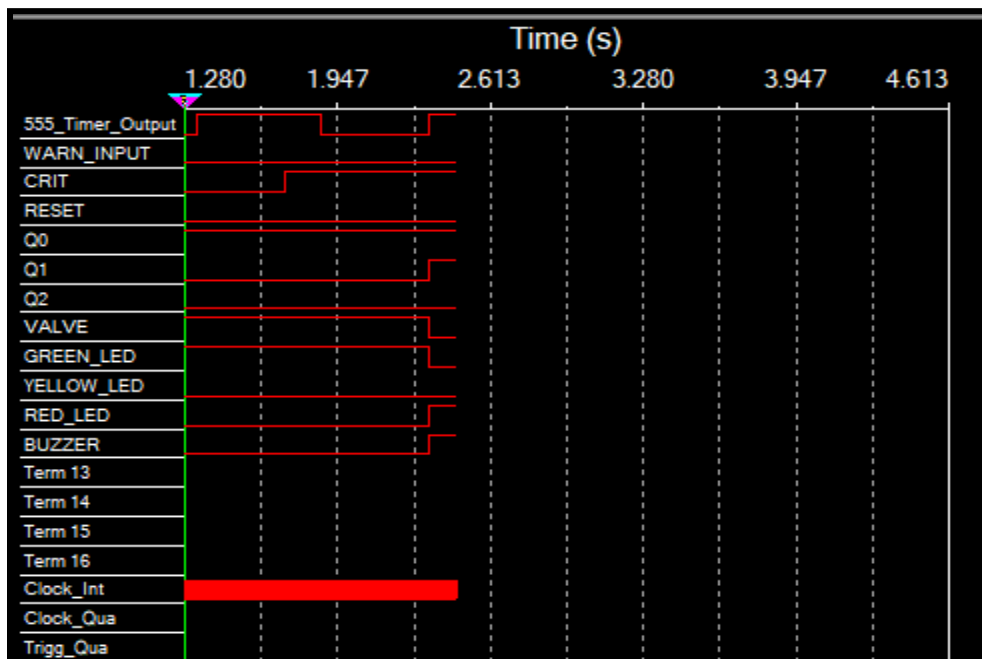
3.6 Clocking Oscillations

- 1.) 000 TO 100 (Normal state to Warning State)



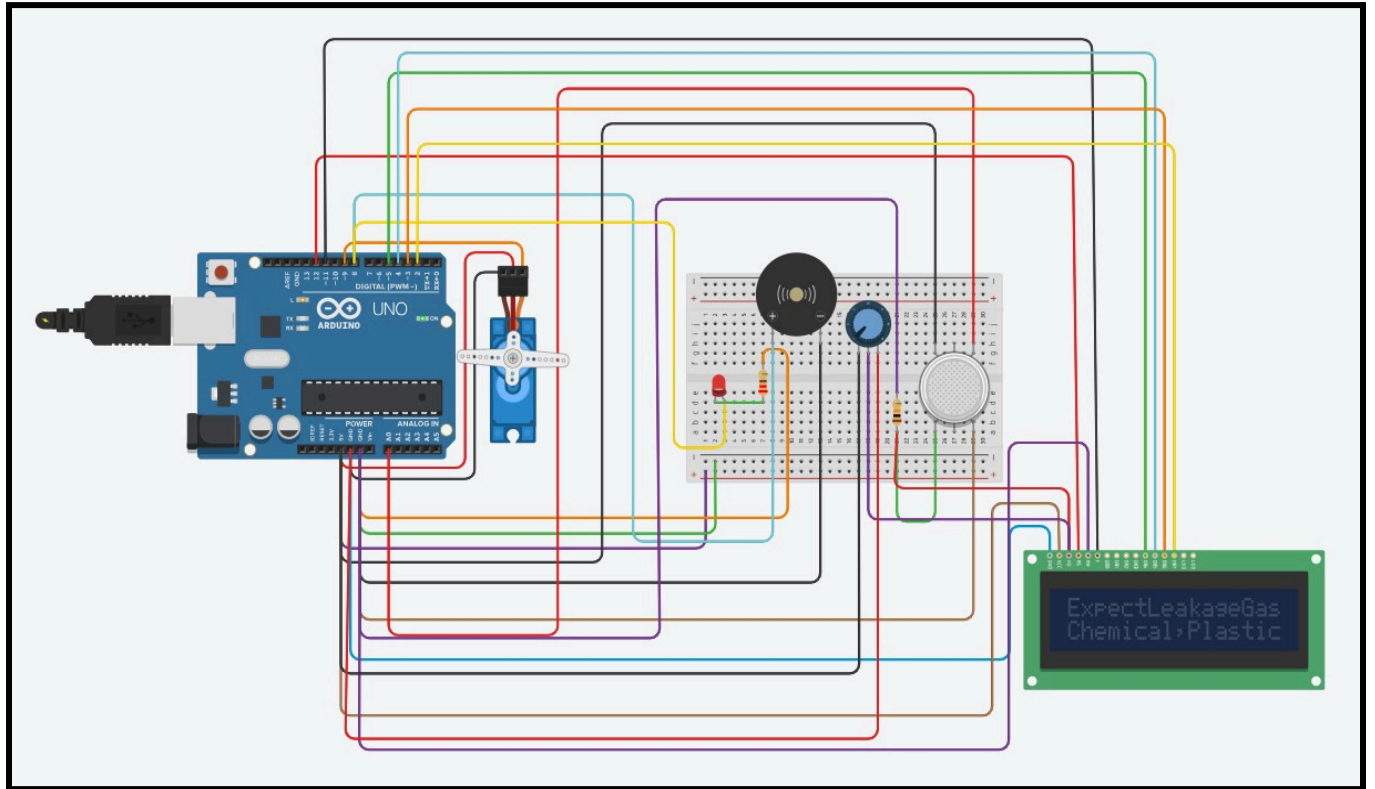
The WARN input was raised high at $t=1.7s$. At the next rising edge of the 555 timer cycle, the green LED signal turns low and the yellow LED signal turns high.

2.) 000 TO 010 (Normal State to Crit State)



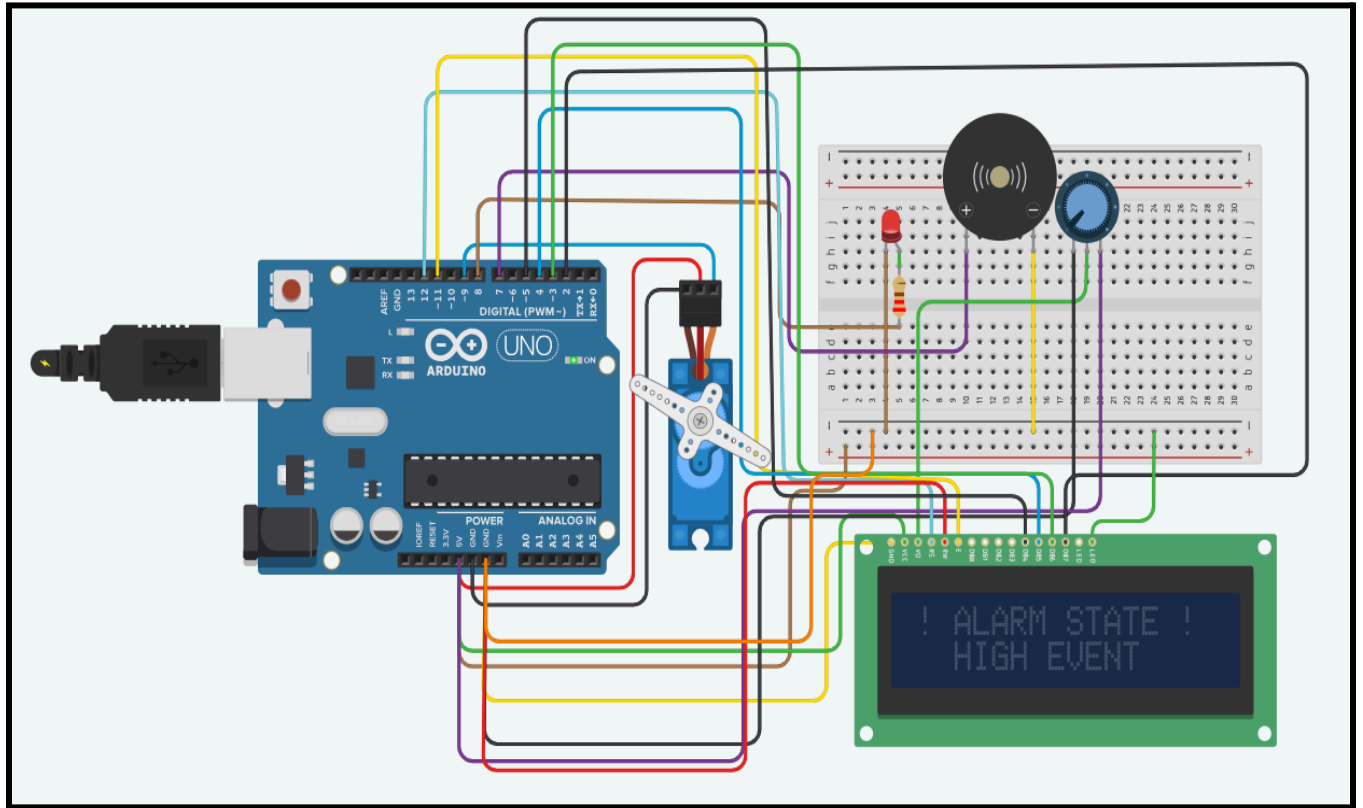
The CRIT input was raised high at $t=1.6$ seconds. At the next rising edge of the 555 timer cycle, the green LED and the valve signal turn low while the red LED and the buzzer signal turn high.

3.7 Power Circuit



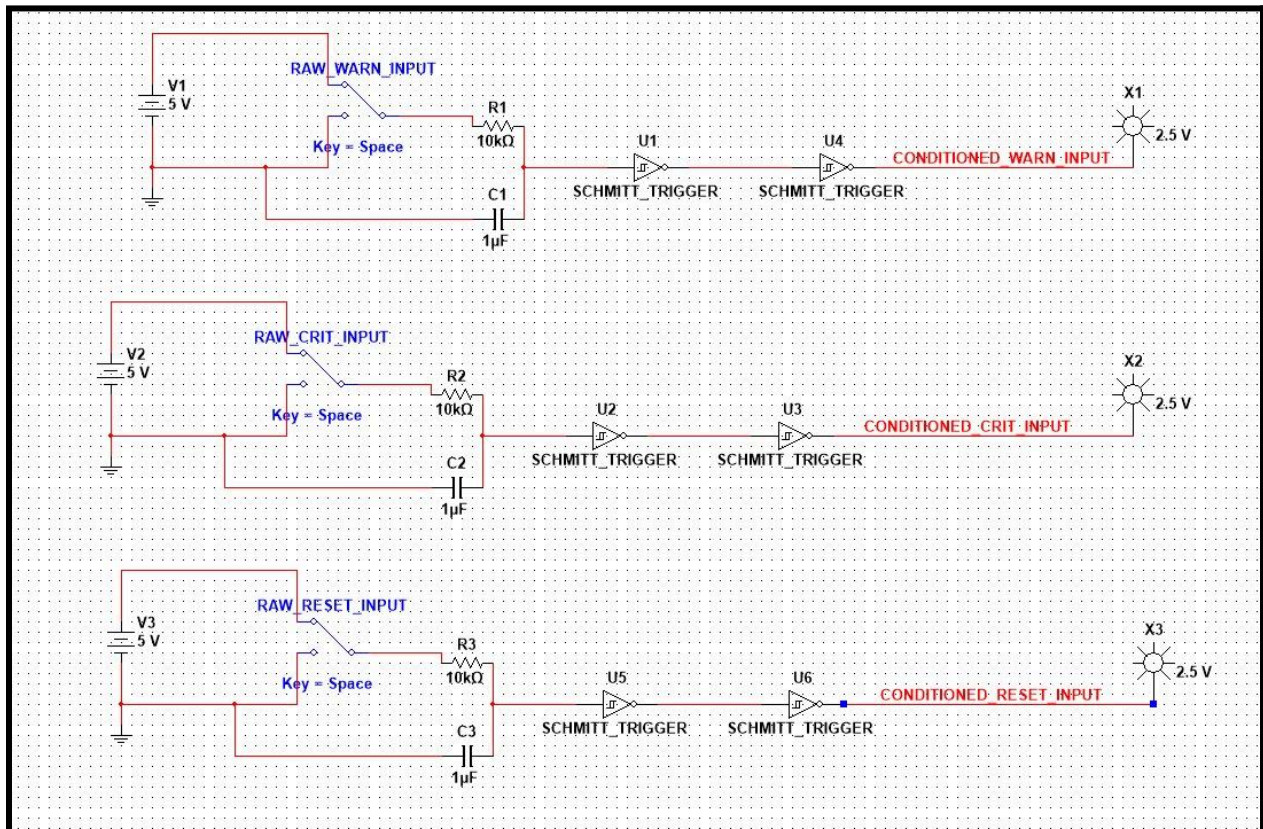
This circuit represents the initial warning state of the prototype, where power is supplied to the main basic output components at its warning state before the system enters the full alarm condition. In this stage, the red LED blinks and the buzzer sounds to indicate that an abnormal or unsafe condition is being expected, while the LCD displays an warning message which is “Expect Leakage Gas, Chemical and Plastic”. It shows a primary warning to users to expect an event of Gas, Chemical and Microplastic Leakage in the surrounding environment by the prototype device and also ensures power is being supplied to the circuit which confirms the basic working methodology of the prototype is working.

3.8 Motor Circuit



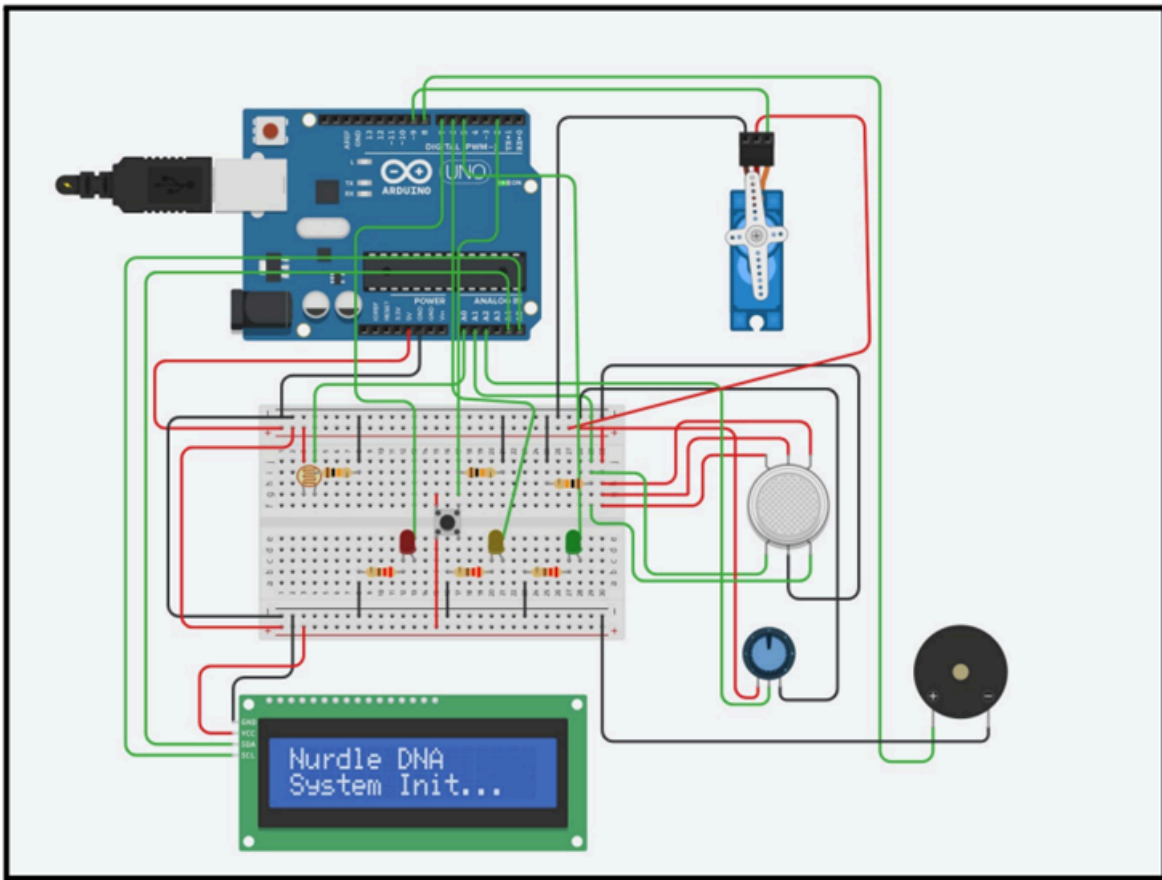
The prototype's motor circuit uses an Arduino-controlled servo motor to produce precise mechanical movement during the alarm state. When a high event is detected, the Arduino sends control signals to activate the servo, while the red LED, buzzer, and LCD message also turn ON to alert the user. This makes the system responsive, accurate, and effective for automatic warning and action.

3.9 Debouncing Circuit



The Debouncing Module acts as a filter that cleans up the signals from the buttons before they reach the FSM. It prevents the input signal from being sent multiple times. To do so, this circuit uses an RC low-pass filter (10kΩ resistor and 1μF capacitor). The Schmitt Trigger chips then turn that smooth signal back into a perfect, sharp square wave for the logic gates to read. This ensures that every button press is counted as exactly one clean action.

4. Sensor Circuit



The sensor circuit consists of the various sensors used in the Nurdle DNA project. The sensors are:

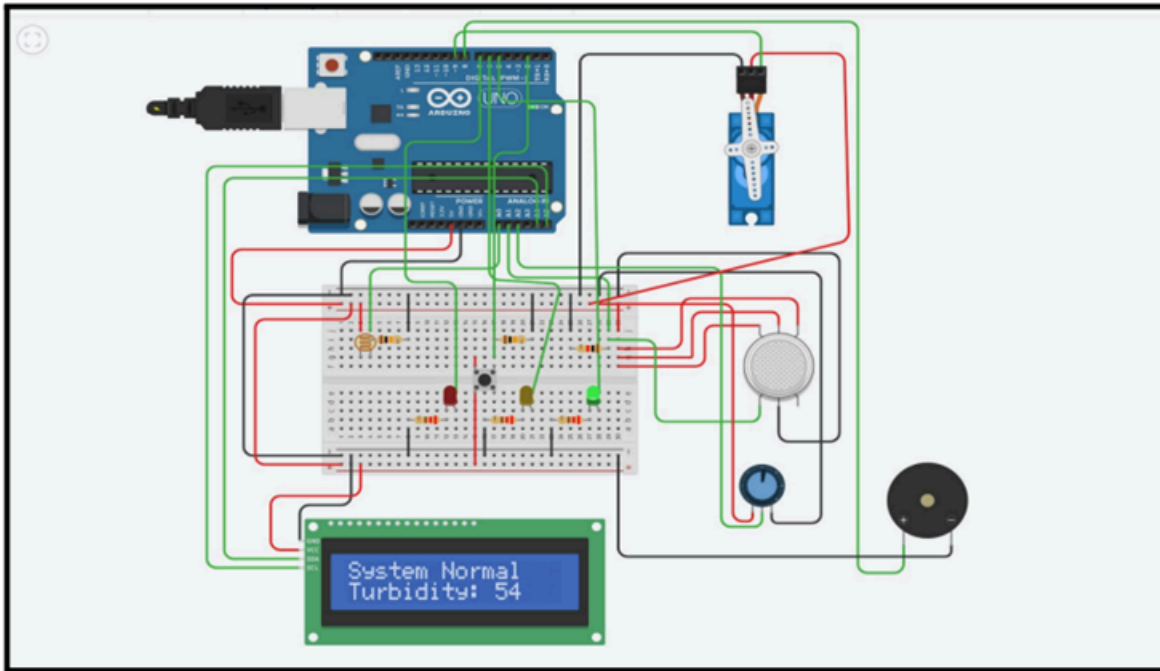
- 1.) A camera (We use a LDR light sensor to simulate the function of a camera in Tinkercad.)
- 2.) Round MQ-style gas sensor
- 3.) A Reset push-button

The circuit also consists of our outputs:

- 1.) 3 LEDs (red, yellow, green)
- 2.) Piezoelectric buzzer
- 3.) Blue servo motor (Valve Control)

Finally, at the bottom, an I2C 16x2 LCD display provides visual feedback.

4.1 Camera (LDR)



The LDR (Light Dependent Resistor) sensor acts as a light-based detection sensor, simulating the Jetson Nano AI camera used in the final project. In this Tinkercad prototype, we manually adjust the sensor's resistance to mimic different levels of turbidity (water cloudiness) caused by microplastics

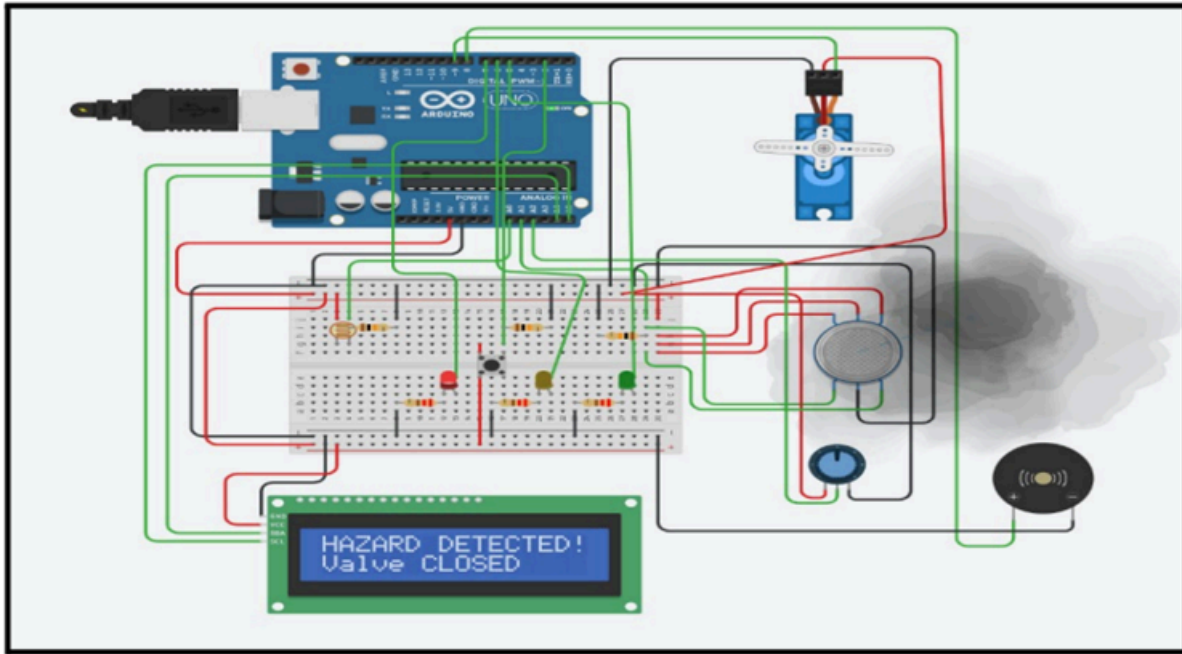
1.) Low Turbidity (Normal State): When the LDR resistance is set to simulate clear water, the green LED stays on. This tells the operator that no nurdles are detected and the valve can remain open.

2.) Moderate Turbidity (Warning State): When the LDR resistance is increased slightly, the system triggers the warning state. The green light swaps to a yellow LED, but the valve stays open to maintain flow

3.) High Turbidity (Critical State): When the resistance is moved to the highest level, it simulates a high density of nurdles. The system immediately

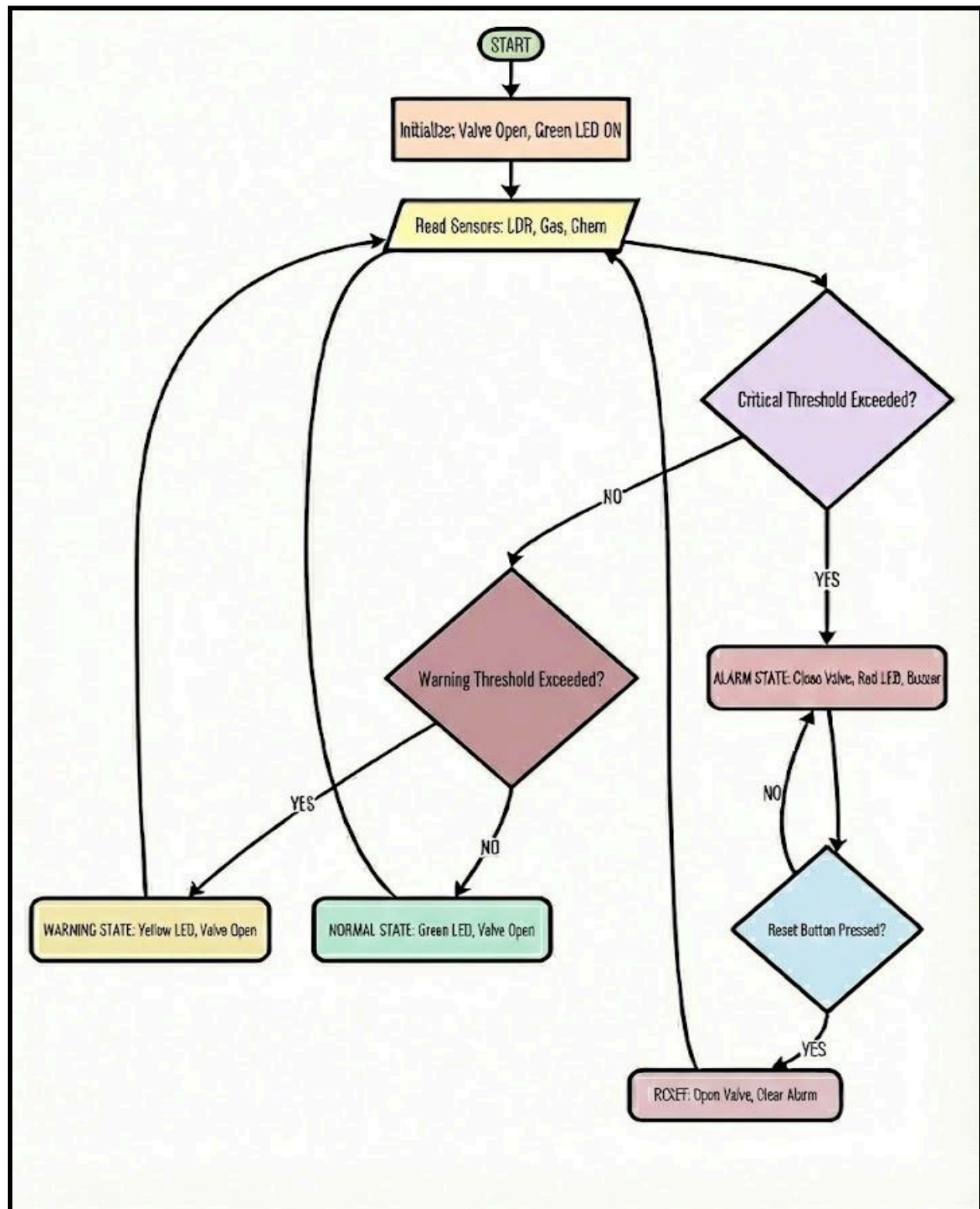
activates the red LED and buzzer, and the servo motor rotates to the closed position to stop the flow of contaminated water.

4.2 Gas sensor

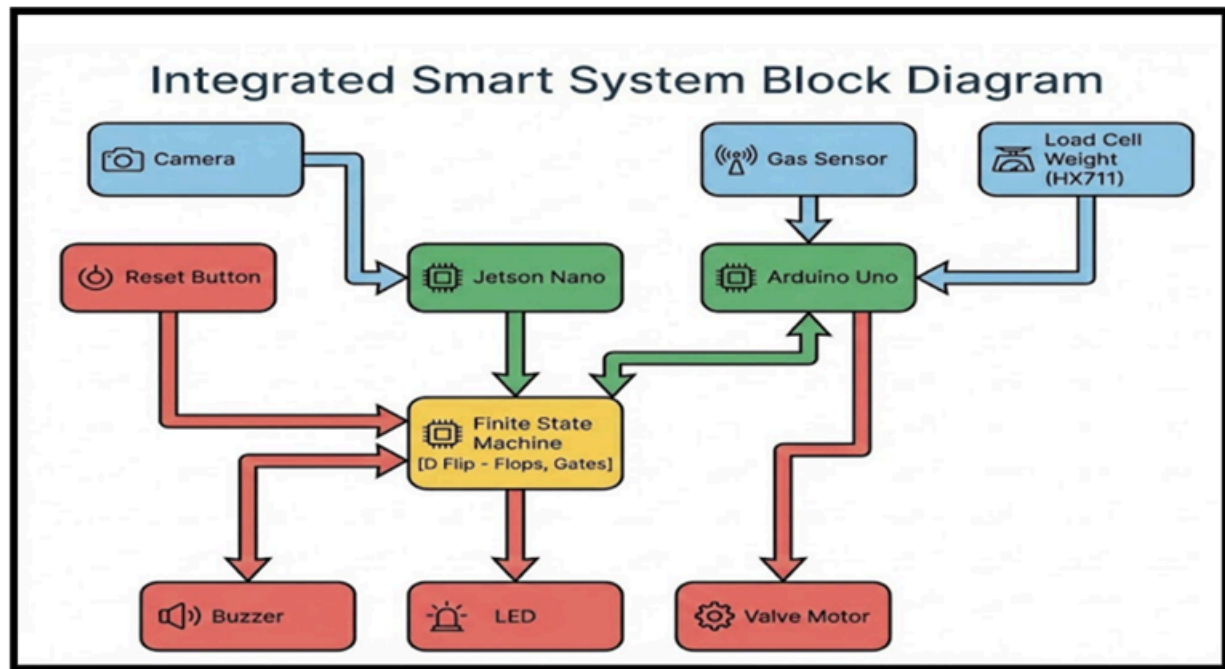


The gas sensor acts as a primary detection interface that monitors for airborne contaminants. In the above test scenario, a plume of smoke near the gas sensor triggers it, prompting the buzzer to emit an active alert signal (The CRIT input becomes 1). In this critical state, the red LED turns on and the blue servo motor rotates into a closed position to simulate an emergency shut-off of the valve, while the LCD screen provides a confirmation of the event: "HAZARD DETECTED! Valve CLOSED."

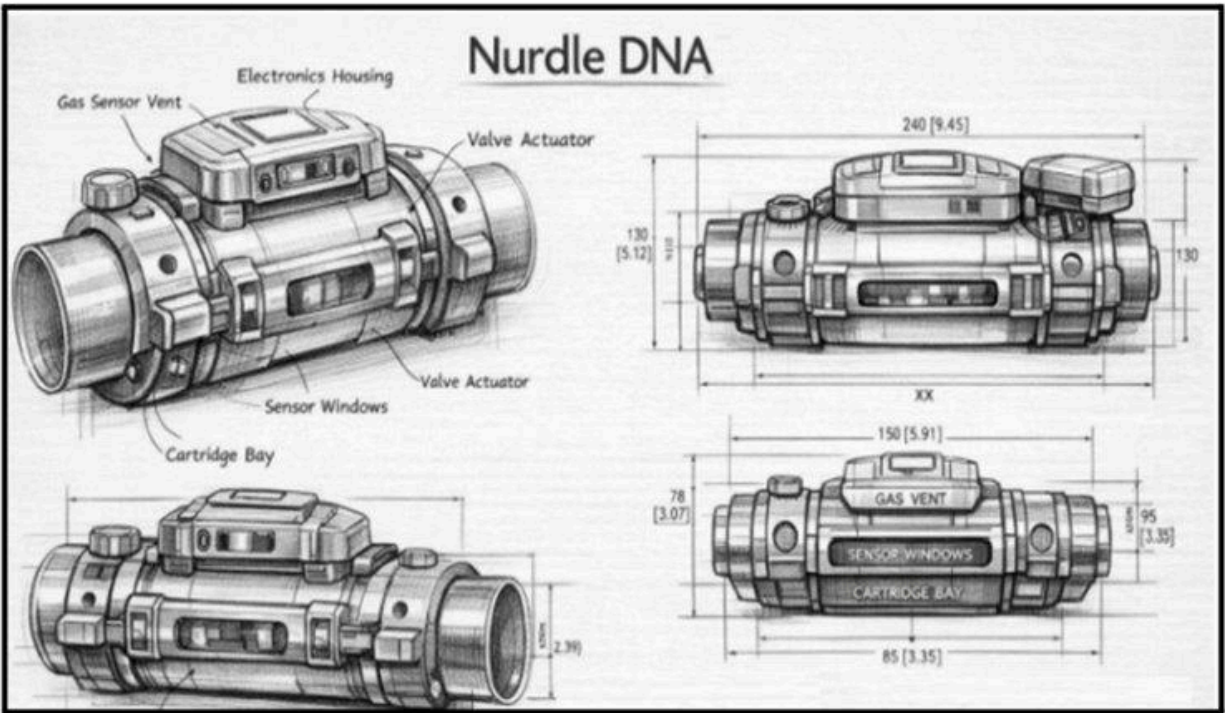
4.3 Flowchart of Sensor Logic



4.4 Block Chart Diagram



5 Prototype Sketch



Prototype 3D Model



6. Feedback

Following the submission of Deliverable 2, our team received valuable feedback on the report. This feedback highlighted several areas for improvement, including technical depth, clarity of explanations, the integration of visual aids, and the overall structure of the report. In preparing Deliverable 3, we have thoroughly reviewed each comment and addressed them systematically throughout this document. The following section details how we have incorporated this feedback into the documentation of different topics of this design report.

State Machine

- Each of the five states (INIT, SysOk, Causn, ALRM, RSTIN) now has a dedicated subsection describing purpose, input conditions, and output configurations
- The transition table clearly shows logical relationships between WARN, CRIT, and RST inputs and next-state decisions
- The complete state diagram on page 18 provides a visual overview of all states with labelled transitions

Clock & Timing Circuits

- Section 2.3 now includes specific component values: $R_1 = 10\text{k}\Omega$, $R_2 = 68\text{k}\Omega$, and $C_1 = 10\mu\text{F}$
- Explains astable mode configuration generating 1Hz clock pulse for synchronous operation
- Clarifies how the clock ensures state machine updates occur once per second

Motor Circuits

- New subsection explains the servo motor's role in converting digital signals to mechanical motion
- Describes a three-wire configuration and Arduino's pulse signals commanding the servo to predefined positions during alarm state

Input Conditioning and Debouncing

- Expanded explanation covers the debouncing purpose (preventing multiple signals from one button press)
- Details about a RC low-pass filter (10k Ω resistor, 1 μ F capacitor) and Schmitt Trigger chips that convert the smoothed signal into a clean digital signal.digital logic
- Ensures every button press registers as exactly one clean action

TinkerCad simulations

- The three simulation screenshots show the system across operating conditions:
 - Initialization: LCD displays "Nurdle DNA System Init"
 - Normal state: Green LED on, LCD shows "System Normal Turbidity: 54"
 - Alarm state: Red LED on, buzzer sounding, servo closed, LCD displays "HAZARD DETECTED! Valve CLOSED" with smoke near gas senso
- Each screenshot includes descriptive text explaining system status and component behaviour
- The flowchart was added to visualise state transitions

These updates directly address all feedback points by providing improved technical depth, clearer explanations, enhanced visual aids, and better overall structure.

7. Conclusion

The design simulation for ECTE250 Deliverable 3 successfully confirmed that the prototype's subsystems work together as intended. The correct outputs, timing behaviour, and state transitions show that the design is reliable and ready to move from simulation to physical prototyping and lab testing.